PRIVATE RESEARCH - NOT INVESTMENT ADVICE

The funding-risk engine, unified scientific report

5th edition - 8 November 2026 - prepared for Tijs de Boer

The short version

1. The recommended mix is now set by two default views instead of one, and judged under five market laws instead of four. Under today's market rates the validated mix holds only 10% in T-bills and scores $120.7M over 10 years; the fitted-history view stays more careful at 32% bills and $109.2M. Both pass every check under both laws.
2. Read in today's dollars the safety promise shrinks. The recommended mix is worth $79.4M in real terms, not $120.7M - and no fixed mix passes your 15% final-loss limit once inflation is priced in. The nominal answer was partly an inflation illusion.
3. The regime law (market memory: calm weeks follow calm weeks) now votes on every candidate. On the fitted view it is the binding law: it pushed the safe share from the 10% floor up to 32% bills. Under current rates it changes nothing - bills at 4.2% carry the portfolio past both laws.
4. The true 10-year optimization is too noisy to trust as a steering wheel. Re-run on fresh paths at three sizes, the risky sleeve swings wildly (US stocks anywhere from a quarter to over half of the risky part). The score barely moves. We keep it as a cross-check, not the default.
5. The model-wrongness margin stays as it was: a real calibration says it should be ~100,000 times larger than the placeholder, and making it real would break every budget. That is a budget decision for you, not a math problem.
6. Clever has still not beaten simple. Across every law, view and horizon tested, no trading rule, objective formula or forecast model beat the plain fixed mix once safety was enforced.

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1. The question

Pay out $250,000 every month for ten years from a $100M balance sheet, without being forced to sell investments at a bad time, and keep as much wealth as possible at the end. The engine answers: which fixed mix of assets gives the best risk-adjusted outcome while passing every safety check?

Input	Value used	Status
Total wealth	$100M	illustrative
Payments	$250k per month, 120 months ($30M)	12 months confirmed; the 10-year extension is a labeled assumption
Horizon	10 years (520 weeks)	your default, 8 Nov 2026
Property	10% of wealth, Singapore residential, not sold, no rent	assumption
Liquid assets	3-month T-bills, IEF (7-10y Treasuries), TIPS, VTI (US stocks), VWO (emerging), GLD (gold), 0% cash account	real funds, real histories
Safety limits	20% per-payment tail allowance at 95%; worst-path loss cap 40%; final-loss cap 15% at the horizon end	your defaults, 8 Nov 2026
Views	two default views: fitted history, and current market quotes (5 Nov 2026)	new in this edition
Judge	five market laws on independent fresh paths	fifth law new in this edition

Two terms: the score is the certainty equivalent - the guaranteed dollar amount after all payments that the model values exactly as much as running the risky plan. A market law is one complete assumption set about how prices move; the final judge now runs five of them so no answer rests on one guess.

2. How the engine now works

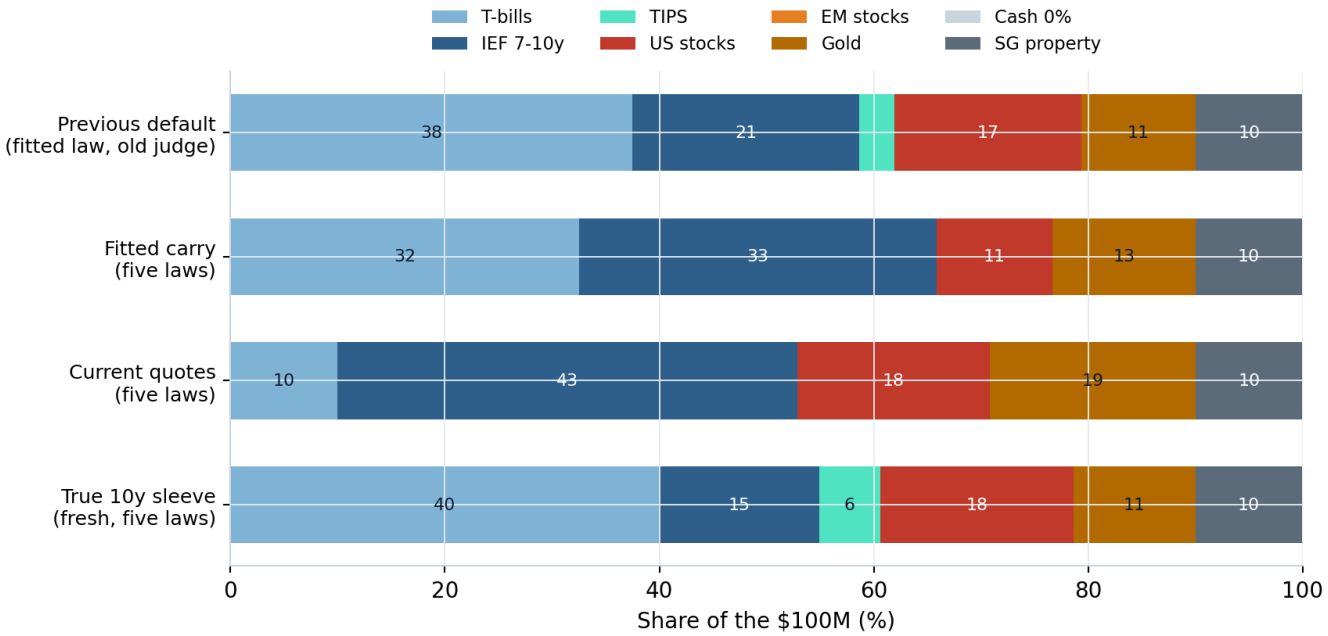
1. Inputs: wealth, the payment calendar, the limits, the fixed property.
2. Estimation: real weekly total-return histories, shrunk toward safe values because raw estimates are noisy. New: the safe assets' carry is estimated twice - once from the full fitted history and once from current market quotes (3M bill 4.17%, 10y 4.84%, 10y TIPS real 2.60%, FRED, 5 Nov 2026). The two answers are reported side by side as the two default views.
3. Scenarios: thousands of simulated 520-week futures for all assets at once.
4. Optimizer: picks the long-only mix with the best score subject to every funding and loss limit.
5. Independent judge: the frozen mix is re-checked on entirely fresh paths under five laws: the four fixed envelopes (reference, nominal, stressed, and their common hull) plus, new in this edition, the regime/bootstrap law - a bootstrap of real historical weeks with volatility that follows recent data. Extra laws only judge; they never fit, solve or repair.
6. Execution audit: a separate cash-first ledger counts whether every payment was actually made.

Everything in this edition was re-run on a fresh seed family: training, validation and final paths are all disjoint from every earlier study. Nothing was tuned on the paths it is judged on.

3. The recommended portfolio

The headline mix - current-quotes view, five laws - holds 10% T-bills, 43% IEF, 18% US stocks, 19% gold and 10% property. It scores $120.7M over ten years on the fitted law and $126.6M when scored by the regime law. Five percent of outcomes end below $106.3M; the median outcome is $159.2M.

Portfolio	T-bills	IEF	TIPS	US stocks	Gold	SG prop.
Previous default (four envelopes, fitted)	37.5%	21.2%	3.2%	17.5%	10.6%	10.0%
Fitted carry view (five laws)	32.5%	33.4%	0.0%	10.8%	13.4%	10.0%
Current-quotes view (five laws)	10.0%	42.9%	0.0%	17.9%	19.2%	10.0%
True 10y sleeve (fresh, five laws)	40.0%	14.9%	5.7%	18.0%	11.4%	10.0%



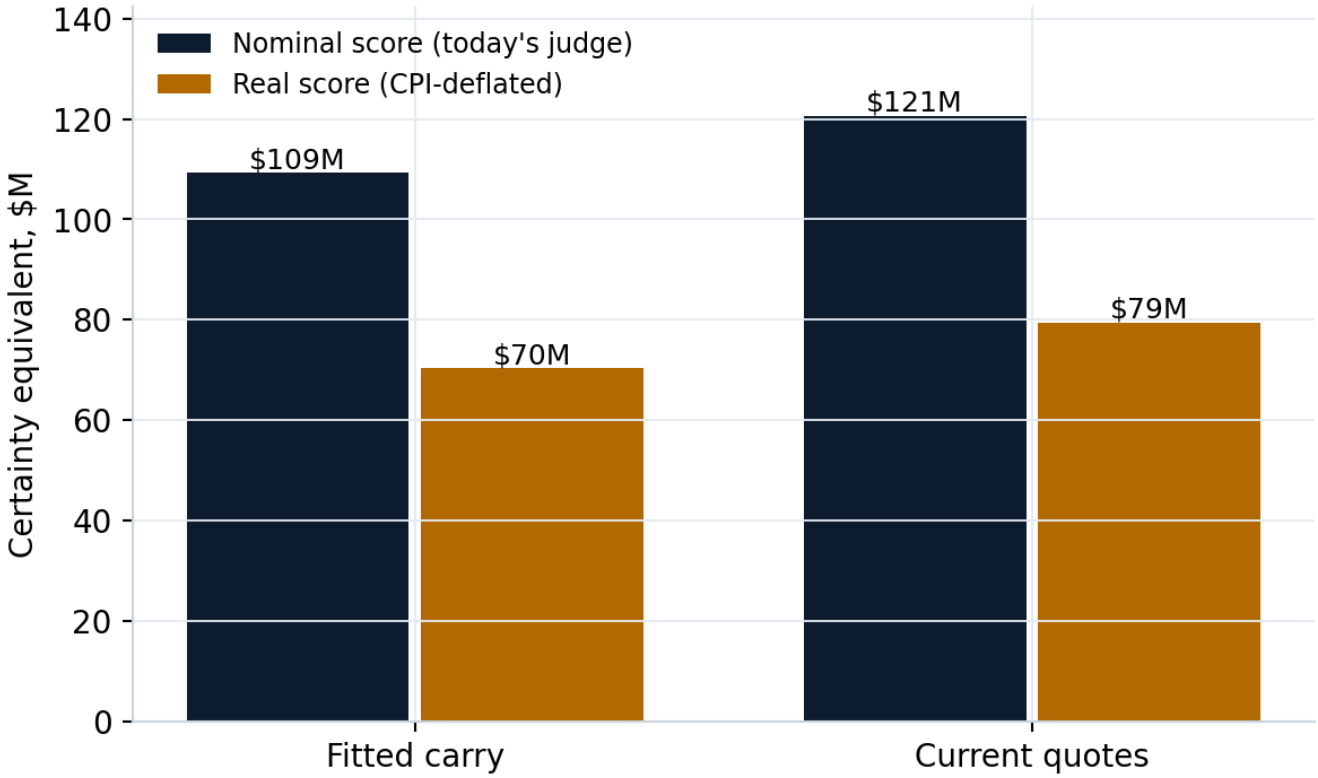
The validated mixes. The two default views disagree about the safe share: the fitted-history view still wants a third in bills; at current quotes the same checks pass at the 10% floor.

The two views disagree, and the disagreement is the point. The fitted history says bills yield ~1.8% and the mix needs 32% of them to pass all five laws (score $109.2M). Today's quotes say bills yield 4.17%, and at that carry the same checks pass at the 10% grid floor with a $120.7M score. The engine now shows both; it does not average them into a false compromise.

Which to hold? If you believe today's bill curve persists, the current-quotes mix dominates on every judge we have, nominal and real. If you distrust the quotes and trust the fitted history, hold the fitted-view mix. The engine's answer is the pair, not a coin flip.

4. Real dollars: the honest headline

Everything so far is nominal: dollars of the year they arrive in. You asked for the real view - today's purchasing power - as the headline. Here it is.



The same two recommended mixes, scored nominally and in CPI-deflated real terms. The gap is the inflation illusion: bills look safe only because the nominal judge never asks what a dollar buys.

In real terms the current-quotes mix is worth $79.4M and the fitted mix $70.3M - against nominal scores of $120.7M and $109.2M. And under the real judge, both mixes fail your 15% final-loss limit at ten years. No fixed mix passes it: not the old sleeve, not a re-solved sleeve, not 90% TIPS. The tightest limits any static mix can meet in real terms are about 45% worst-path and 40% final (90% TIPS plus the fixed 10% property, real score $80.5M).

- How real terms are built: every scenario is deflated by a bootstrapped inflation path (US CPI, 1947-2026, mean 3.5% a year), payments become inflation-indexed (a constant real $250k/month), and TIPS are modeled as a genuinely real asset at the live 2.60% real yield.
- Inflation paths are drawn independently of market paths - a labeled simplification; TIPS get their inflation link mechanically by construction.
- Your standing defaults stay nominal (40% worst-path, 15% final at each horizon's end). The real view is a re-pricing of the same portfolios, reported alongside, not a new set of limits - changing the limits is your call, and section 9 prices it.

5. The regime law joins the judge

Markets have memory in their volatility: calm weeks follow calm weeks, storms follow storms. The default law ignores this - every week is drawn independently. Earlier editions flagged it as the biggest model risk. This edition does something about it: the engine's bootstrap/regime variant (EWMA-filtered resampling of real historical weeks) is now a fifth judging law. Every candidate must pass on independent draws of both laws.

- On the fitted view the regime law binds: the safe share rises from the 10% floor to 32% bills, costing $2.5M of score against the single-law answer.
- On the current-quotes view it changes nothing: bill carry at 4.17% passes both laws at the floor. The regime law even scores that mix higher ($126.6M) than the default law does - its historical weeks include long stretches of high bill rates.
- Judged alone, the regime law is kinder than the default (its frozen-sleeve answer: 10% bills, score $123.3M). It is not a harsher world; it is a different one. That is why it belongs in the judge.

The regime law's own caveat stands: as wired, it is a current-volatility-regime law - its resampled weeks are scaled by today's volatility level, so it under-weights calmer regimes than today's. That makes it a complement to the default law, not a replacement.

The policy-by-model matrix, frozen

Policy (all five-law validated where shown)	Default law	Regime law	Real terms
1y sleeve, fitted carry (recommended alt.)	$109.2M	$113.1M	$70.3M
1y sleeve, current quotes (recommended)	$120.7M	$126.6M	$79.4M
True 10y sleeve (fresh)	$109.6M	$116.3M	-
True 10y sleeve (fix 4, on fresh paths)	$112.2M	$119.4M	-
Previous default (pre-round-2)	$110.8M	-	$71.6M

Scores are certainty equivalents over 10 years on shared frozen paths; each column judges under a different law, so columns are not utilities compared - they are the same CARA score under different worlds. Real terms deflate by bootstrapped CPI. Missing cells: not yet computed on this seed family, not hidden failures.

No policy wins everywhere, and no complex policy beats the simple fixed mixes: the two 1y-sleeve mixes and the two 10y-sleeve mixes sit within a few $M of each other under every law, and the regime law actually prefers the riskier current-quotes mix. The simple static policy remains the answer; the matrix is the proof, frozen.

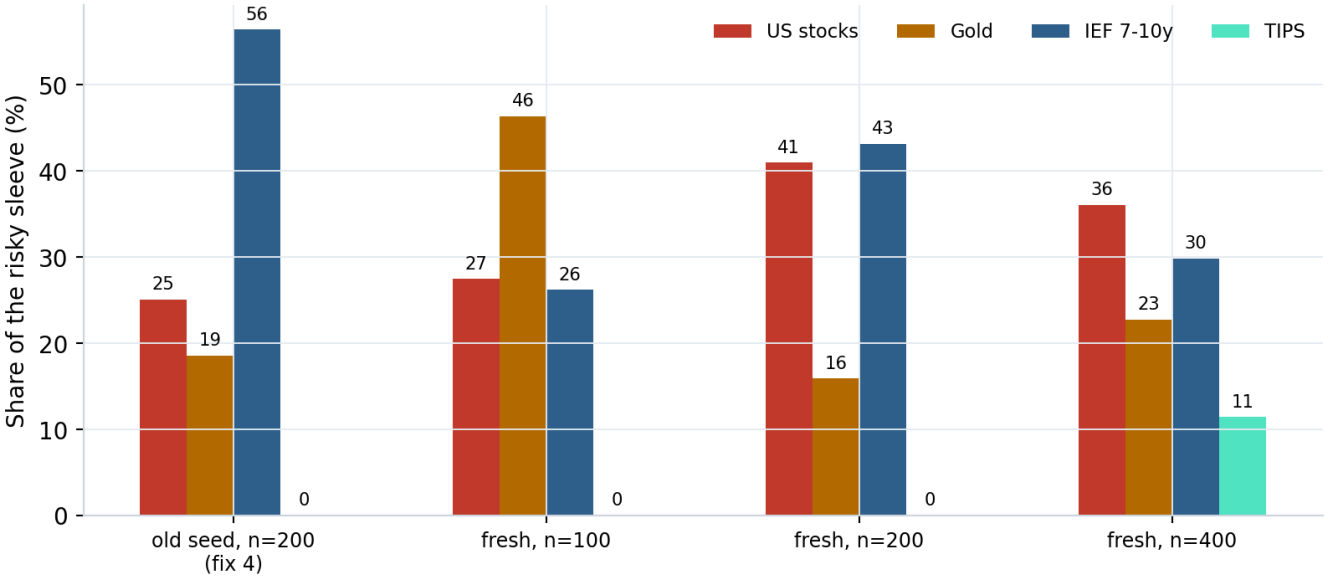
6. The second default view: current market rates

The fitted estimation window quietly drove the old answer: it has bills earning ~1.8% because that is the shrunk historical average, while a 3-month bill pays 4.17% today. The engine now always computes both views. Fitted means: score $110.8M at 38% bills (old judge) / $109.2M at 32% bills (five laws). Current quotes: $120.7M at 10% bills.

- What changes in the quotes view: only the safe assets' expected carry (bill 4.17%, IEF 4.84%, TIPS 2.60% real + 2.24% breakeven). Volatilities, correlations and all risky-asset means stay fitted. Labeled simplification: IEF/TIP expected return is set to the current yield, with no roll or term-premium model.
- Quotes are a dated snapshot (FRED, 5 Nov 2026), not a maintained feed. The view goes stale; the date travels with it wherever it is shown.
- The quotes view also holds up under the regime law and in real terms ($79.4M real vs $70.3M for the fitted mix).

7. The ten-year optimization, properly

The old pipeline solved the risky sleeve on a one-year problem and stretched it to ten years. The fix round solved the true 10-year problem directly. This edition re-ran that solve on completely fresh paths at three scenario sizes - and the honest finding is that the true 10-year solve is too noisy to steer by.



The risky sleeve from the true 10-year solve, by scenario count and seed family. The proportions swing far more than the score does: the optimum is flat, so the solver chases noise.

- Fresh solves: n=100 in 21s, n=200 in 52s, n=400 in 696s

- Sleeve instability: the risky-sleeve proportions move by 0.32 (L1) between the old and fresh seed at n=200 - the solver reallocates most of the risky part between runs while the score moves by about a million dollars.
- Put through the five-law discipline anyway, the fresh 10y sleeve validates at 40.0% bills and scores $109.6M
- The old-seed 10y sleeve, re-judged on genuinely fresh paths, validates at 15.0% bills and scores $112.2M - the fix-4 result survives out-of-sample
- Verdict: keep the 1y sleeve as the default; the 10y solve stays a cross-check. The score difference between them is small and unstable in sign.

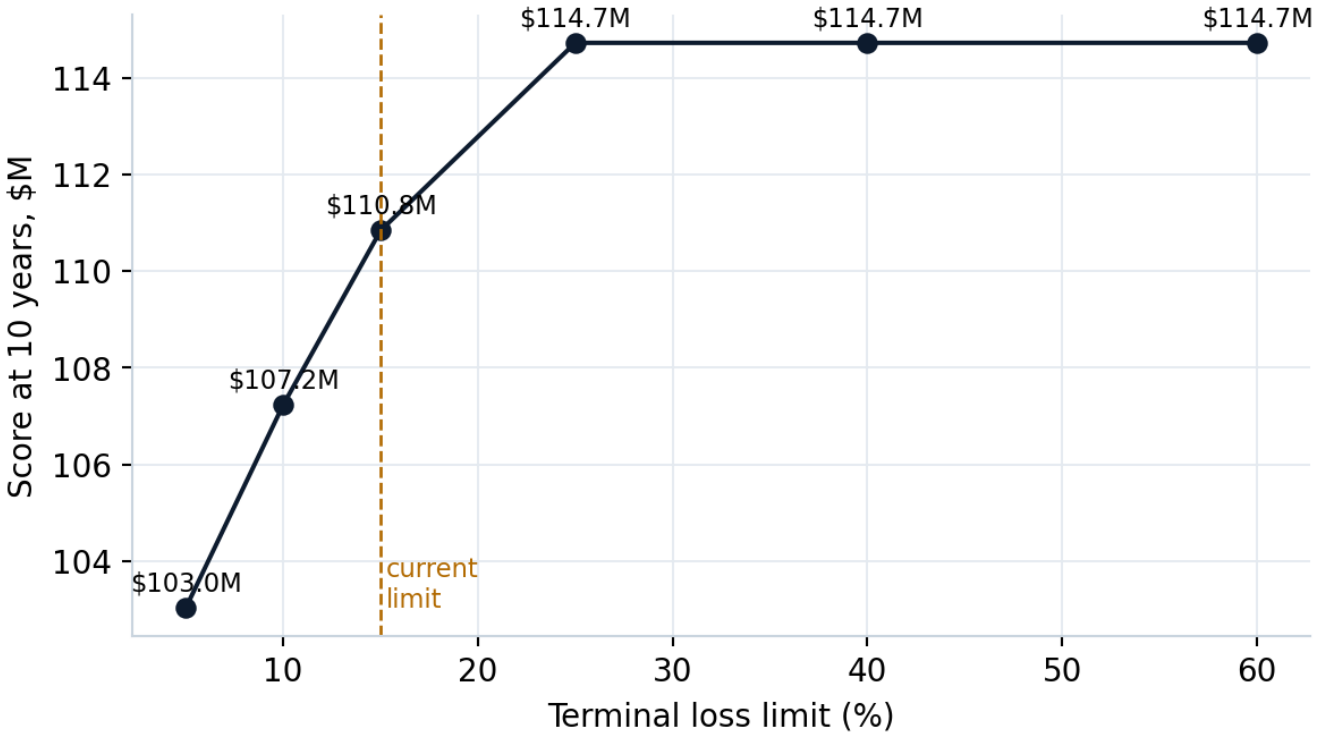
8. The model-wrongness margin: still a budget decision

The engine keeps a safety margin for the market model itself being wrong (the Wasserstein radius). The placeholder is 0.00002. Calibrated from how much the fitted model actually moves across refits, it should be about 2.55 - roughly 100,000 times larger. Nothing in this edition changes that math.

- At the placeholder, the margin already spends about $40k of the $50k smallest payment budget.
- One step up (0.000025) the current portfolio still passes; two steps (0.00003) and no mix is feasible at any setting. A statistically real margin is unaffordable under the 20% per-payment budget.
- That is why this stays open: it is not a calibration problem, it is a budget decision - how much payment flexibility you are willing to spend on model wrongness. Your plan: leave it, documented.

9. What the loss limits cost

At the standing 40% worst-path cap, tightening the final-loss limit from the loosest tested setting to 15% costs $3.9M of score; to 10% costs $7.5M; to 5% costs $11.7M. Loosening 15% to 25% recovers the full $3.9M - and nothing beyond that, because the 40% worst-path cap starts binding instead.



Score versus the final-loss limit at the standing 40% worst-path cap. The current 15% sits on the cheap part of the curve; 10% and 5% get expensive fast.

The same question in real terms has a harsher answer (section 4): at 15% real, nothing passes. If you ever want the limits stated in real terms, the feasible pair is about 45% worst-path / 40% final.

10. What carries over from earlier editions

- 19 frozen annual holdouts of real history, matched objectives, 152 historical outer refits and the payout stresses: no objective formula, forecast model or trading rule beat the plain fixed mix once safety was enforced. (4th edition, sections 5-9.)
- Rebalance rarely: annual scored best of the tested frequencies; at one year that means set the mix once and do not trade. No frequency was accepted across all laws at 3-5 years under unchanged budgets - reported, not hidden.
- Utilities: CARA, CRRA, HARA and the kinked-shortfall objective pick the same portfolio at the confirmed $250k/month. The answer does not depend on the utility flavor.
- Payouts $50k/$100k/$250k per month: the validated mix barely changes; the loss caps, not the payments, set the safe share.
- Execution audit: the cash-first ledger pays every payment in the validated configurations; the strict 99% all-payments check remains a separate, stronger line.
- Regression: 380 tests pass against this edition's engine changes (second view, fifth law, review-brief fixes). Two pre-existing failures are unchanged from the base commit - hardcoded paths in two committed study configs and one solver accuracy flag in this sandbox; neither touches the engine.

11. Everything that is still open

- Real vs nominal limits: the one decision that shapes everything else. This edition shows both; it does not choose for you.
- The model-wrongness margin (section 8): a budget decision, parked.
- The regime law is a current-volatility-regime law as wired; a multi-regime version is research, not a patch.
- The 10-year optimizer needs scenario counts beyond this machine's budget before its sleeve can be trusted as a default (section 7).
- The client is still a placeholder: real SFO inputs (actual obligations, tax, other accounts) would change the answer more than any modeling choice in this report.
- The current-quotes view is a 5 Nov 2026 snapshot; it should be refreshed when quotes move.
- Property is a price index, not a tradable total return; no rent, no sale, fixed 10%.
- Data rights: Yahoo-sourced histories are fine for private research; not cleared for sharing.

12. The verdict

Hold the current-quotes mix - 10% T-bills, 43% IEF, 18% US stocks, 19% gold, 10% property - if you accept today's bill curve as the carry you actually earn. It passes all five laws on fresh paths, scores $120.7M nominal and $79.4M real, and it is the mix the engine now shows as its second default view.

Hold the fitted-view mix (32% bills) if you distrust the quotes. It is the conservative answer and it also passes all five laws. What you should not do is average the two: the disagreement is information, not noise.

And read the real-terms headline before spending the nominal one: in today's dollars the recommended mix promises about four-fifths of its nominal score, and the 15% final-loss limit is not achievable by any fixed mix. That is the boundary of what a static portfolio can promise - the engine now says so in both currencies.

Private research for Tijs de Boer, not investment advice. Every portfolio here is a frozen candidate judged on paths it never saw; passing is a statement about the model, not a guarantee about the future.

Appendix A. Data, rights and reproducibility

- Universe: VTI, VWO, GLD, IEF, TIP total-return histories (Yahoo, private use accepted); 3-month T-bill roll built from FRED DGS3MO; Singapore URA private residential index (data.gov.sg); SGD/USD from FRED; CPI from FRED CPIAUCSL (1947-2026).
- Current quotes: FRED H.15 daily, 5 Nov 2026 - DGS3MO 4.17%, DGS10 4.84%, DFII10 2.60%; breakeven 2.24% by subtraction.
- Every study in this edition carries its config, seeds and data hashes in studies/round2/results/ and studies/fixes/results/. Draws are cached and deterministic in (config, seed); the fresh family is seed 20261108, disjoint from the earlier 20261107 family.
- Engine changes in this edition: emfin_v2/current_rates.py (second view), emfin_v2/multi_law.py (fifth law), default config gains secondary_carry_view and validation_laws. Regression suite and smoke test re-run.
- Branch: engine-round2-20261108 on the private repo. No upstream writes; inherited code rights still unresolved for anything beyond private use.

Appendix B. The five laws, precisely

Law	What it is	Role
1. Reference	Empirical distribution of the simulated paths	baseline judge
2. Nominal	Entropy-pooled posterior matching the configured views	judge under your views
3. Stressed	Nominal pushed to a worse equity-basket path loss	adverse judge
4. Common hull	Worst common-threshold bound over the finite hull of 1-3	envelope of the three
5. Regime/bootstrap	Independent draws of the EWMA filtered historical simulation law, judged under its own 1-4 envelopes	new in this edition

Appendix C. Study inventory for this edition

- studies/round2/study_a_dual_law.py: both carry views x single/dual-law mix searches, fresh seeds.
- studies/round2/study_c_longhorizon.py: true 10y solves at n=100/200/400 on fresh paths + stability.
- studies/round2/study_c2_longhorizon_mix.py: fresh and old 10y sleeves through the five-law discipline.
- studies/round2/study_d_real.py: real-terms repricing of the recommended mixes (fix-3 machinery, fresh seeds).
- studies/fixes/study_*.py (fixes 1-6): market models, radius calibration, real terms, true 10y, current rates, constraint grid - results carried into sections 4-9.